# LEVEL UP ACADEMY

Class 10 | GSEB | Science

## Ch 3 Test (GSEB pattern): Metals and Non-metals

*Chapter / Topic: Metals and Non-metals*

Name: _____ Roll No.: _____ Date: _____

**Time: 50 minutes** **Max. Marks: 25**

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**General instructions:**

1. All questions are compulsory.
2. Section A: objective questions, 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 4 marks.
3. Write balanced equations and draw neat, labelled diagrams wherever needed.

### SECTION A

9 marks

*Choose the correct option.*

**Q1.** Which metal is a liquid at room temperature? **[1]**

(a) Mercury (b) Sodium
(c) Gallium (d) Bromine

**Q2.** Which non-metal is a liquid at room temperature? **[1]**

(a) Iodine (b) Mercury
(c) Sulphur (d) Bromine

**Q3.** Which of the following can be beaten into thin sheets? **[1]**

(a) Phosphorus (b) Zinc
(c) Sulphur (d) Oxygen

**Q4.** The property of metals that allows them to be drawn into thin wires is: **[1]**

(a) malleability (b) sonority
(c) ductility (d) lustre

**Q5.** Which metal is stored under kerosene? **[1]**

(a) Iron (b) Copper
(c) Sodium (d) Aluminium

**Q6.** Which of these metals reacts only with steam, not with cold or hot water? **[1]**

(a) Sodium (b) Iron
(c) Potassium (d) Calcium

**Q7.** A metal oxide that reacts with both acids and bases to form salt and water is called: **[1]**

(a) a basic oxide (b) an acidic oxide
(c) a neutral oxide (d) an amphoteric oxide

**Q8.** In electrolytic refining, the anode mud consists of: **[1]**

(a) insoluble impurities (b) soluble impurities
(c) the pure metal (d) the electrolyte

**Q9.** Ionic compounds conduct electricity: **[1]**

(a) in the solid state
(b) never
(c) in the molten state or in aqueous solution
(d) only in the gaseous state

## SECTION B

6 marks

**Q10.** Zinc gives hydrogen gas with dilute hydrochloric acid, but copper does not. Why? Write the balanced equation for the reaction of zinc. **[2]**

**Q11.** Why is sodium kept immersed in kerosene oil? **[2]**

**Q12.** What are amphoteric oxides? Give two examples. **[2]**

## SECTION C

6 marks

**Q13.** Answer the following about the metals aluminium, copper, iron and zinc: **[3]**

**(i)** Arrange them in decreasing order of reactivity. **[1]**

**(ii)** Will a reaction take place when copper is added to iron(II) sulphate solution? Give a reason. **[1]**

**(iii)** Write the balanced equation for the reaction of zinc with copper sulphate solution. **[1]**

**Q14.** Element X has atomic number 12, and element Y has atomic number 17. **[3]**

**(i)** Write the electronic configuration of X and Y. **[1]**

**(ii)** Write the formula of the compound formed between X and Y. **[1]**

**(iii)** Name the type of bond, and give two properties of this compound. **[1]**

## SECTION D

4 marks

*Read the passage and answer the questions that follow.*

**Q15.** Pure gold, known as 24 carat gold, is very soft and is not suitable for making jewellery. It is alloyed with silver or copper to make it hard. In India, jewellery is usually made of 22 carat gold. An alloy is a homogeneous mixture of two or more metals, or of a metal and a non-metal. Brass is an alloy of copper and zinc, bronze of copper and tin, and solder of lead and tin. The melting point and electrical conductivity of an alloy are lower than those of the pure metals. **[4]**

**(i)** What does 22 carat gold mean? **[1]**

**(ii)** Name the metals in bronze. **[1]**

**(iii)** Why is solder used for welding electrical wires together? Give one more reason for making alloys. **[2]**

*~ All the best! ~*